

SQL Subquery Review -Oracle SQL

Query 1

```
SELECT first_name, salary
FROM employees
WHERE salary > AVG(salary);
```

Error: ORA-00934: group function is not allowed here **Reason:** Aggregate functions can't be used in WHERE.

Fix:

```
SELECT first_name, salary
FROM employees
WHERE salary > (SELECT AVG(salary) FROM employees);
```

Query 2

```
SELECT department_name
FROM departments
WHERE department_id = (
    SELECT department_id
    FROM employees
);
```

Error: ORA-01427: single-row subquery returns more than one row **Reason:** = expects one value, but subquery returns many rows.

Fix:

```
SELECT department_name
FROM departments
WHERE department_id IN (
    SELECT department_id
    FROM employees
);
```

Query 3

```
SELECT *
FROM employees
WHERE department_id IN (
    SELECT salary
    FROM employees
);
```

No syntax error, logically wrong - comparing department_id to salary makes no sense.

Fix :

```
SELECT *
FROM employees
WHERE department_id IN (
    SELECT department_id
    FROM employees
    WHERE salary > 10000
);
```

Query 4

```
SELECT first_name
FROM employees
WHERE salary > (
    SELECT salary
    FROM employees
    WHERE department_id = 90
);
```

Risk: Fails with ORA-01427 if department 90 has more than one employee.

Fix:

```
SELECT first_name
FROM employees
WHERE salary > (
    SELECT MAX(salary)
    FROM employees
    WHERE department_id = 90
);
```

Query Logic Analysis

Query	Subquery Type Needed	Result Rows	Correct Operator	Execution Order Issue	Business Logic
1	Scalar	Single-row	>	Subquery must run first to give one value	Correct after fix
2	Multi-row	Multi-row	IN	Subquery returns a list	Correct after fix
3	Multi-row	Multi-row	IN	Runs fine, wrong columns compared	Needs redefinition
4	Scalar (aggregate)	Must force single-row	>	Subquery can return multiple rows	Correct after fix

Final Reflection

- **Subqueries in HR reporting:** Allow dynamic comparisons (e.g., vs. average/max salary) without hardcoding values.
- **JOIN vs Subquery thinking:** JOIN combines tables into wider rows; subquery produces a value/list used as a filter condition.
- **Why runtime errors happen:** Single-value operators (=, >) used with subqueries that return multiple rows, or aggregates used in `WHERE`.
- **Why execution order matters:** Subquery runs first and its result determines what the outer query compares against.
- **Why logic > syntax:** A query can run with no errors but return meaningless results if the logic doesn't match the actual business need.